

The Birth of Quantum Mechanics

1, 3 September • Objectives 1, 2

In eighteen months, 1925–1926, two incompatible-looking formalisms appeared and were proven equivalent. What emerged was a theory with a deterministic evolution law, a probabilistic measurement rule, and no agreed account of how the second relates to the first. That gap is the measurement problem, and it is still open.

Two formalisms

Heisenberg, 1925. Working from the position that a theory should reference only observable quantities — spectral frequencies and intensities, not unobservable orbits — Heisenberg built an algebra of transition amplitudes. Born recognized the arrays as matrices. The defining result is that they do not commute:

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

Schrödinger, 1926. Following de Broglie's standing waves, Schrödinger wrote an equation for a continuous wavefunction $\psi(x, t)$:

$$i\hbar \partial\psi/\partial t = \hat{H}\psi \qquad \hat{H} = -(\hbar^2/2m)\nabla^2 + V(x)$$

Schrödinger proved the two equivalent in 1926: same predictions, different bases. Dirac and von Neumann then showed both are representations of vectors and operators on Hilbert space.

The systems reading: Heisenberg wrote the **operator interface** — what you can do to a state. Schrödinger wrote the **state representation** — what a state is. Neither is more fundamental; they are the same abstraction viewed from opposite sides, which is why Dirac notation, designed to be basis-agnostic, superseded both.

Non-commutation and uncertainty

$[\hat{x}, \hat{p}] = i\hbar$ is the theory's structural core. Two observables whose operators fail to commute have **no shared eigenbasis** — no state assigns both a definite value.

$$\sigma_x \sigma_p \geq \hbar/2$$

This is not measurement clumsiness. It is a statement about the state itself: a state sharply peaked in position is necessarily a broad superposition in momentum, because position and momentum bases are Fourier transforms of one another. The bound is a property of Fourier pairs, and it holds before anyone measures anything.

Analogy boundary — non-commuting operations. Order-dependence is familiar: rotations in 3-D don't commute, and neither do two writes to shared state. The analogy captures order-dependence and misses the deeper claim. Non-commuting matrix operations still act on a state with definite entries. Quantum non-commutation says **there is no assignment of simultaneous values at all**. Bell's theorem in Week 4 makes this precise and testable — which is what elevates it above interpretive preference.

The Born rule

Schrödinger initially read $|\psi|^2$ as physical charge density — the electron literally smeared out. This fails: wavepackets spread, and detectors always fire at points.

Born's 1926 correction, delivered in a footnote:

$$P(\text{outcome } m) = |\langle m | \psi \rangle|^2$$

$|\psi|^2$ is a **probability density**. The state does not tell you what will happen; it gives you a distribution over what can.

This creates the theory's central asymmetry — two evolution rules that do not resemble each other:

UNITARY EVOLUTION (Schrödinger) Deterministic · Linear · Reversible · Continuous $ \psi(t)\rangle = U(t) \psi(0)\rangle, \quad U^\dagger U = I$
MEASUREMENT (Born + projection) Probabilistic · Non-linear · Irreversible · Discontinuous $ \psi\rangle \rightarrow m\rangle$ with probability $ \langle m \psi \rangle ^2$

Every interpretive dispute in quantum mechanics is an argument about the relationship between those two boxes.

Analogy boundary — Born rule as a typed outcome envelope. Reading $|\psi\rangle$ as `Distribution<Outcome>` with weights is a good working model: the type carries possible results and their probabilities. The boundary is that a probability distribution is **basis-independent** — a die's outcome distribution does not change because you ask about it differently. $|\psi\rangle$ yields a different distribution in every measurement basis, and those distributions are not marginals of any joint distribution. That is exactly what Bell's theorem rules out.

The measurement problem

State it as a contradiction between three individually plausible claims:

- 1. The wavefunction is a complete description of a physical system.
- 2. The wavefunction always evolves unitarily (linearly, deterministically).
- 3. Measurements have single definite outcomes.

Take a qubit in $(|0\rangle + |1\rangle)/\sqrt{2}$ and a measuring device D . Linearity gives:

$$(|0\rangle + |1\rangle)/\sqrt{2} \otimes |D_ready\rangle \rightarrow (|0\rangle|D_0\rangle + |1\rangle|D_1\rangle)/\sqrt{2}$$

The device ends in superposition. Nothing in unitary evolution selects a branch. Yet the pointer reads one value. The three claims cannot all be true, and each major interpretation denies a different one:

INTERPRETATION	DENIES	COST
Copenhagen	(1) — ψ is knowledge, not a full description	Where is the classical/quantum cut?
Many-worlds	(3) — all outcomes occur, in branches	Why does the Born rule hold?
Bohmian mechanics	(1) — particles have definite positions too	Explicit non-locality
Objective collapse (GRW)	(2) — collapse is a real physical process	New physics; testable, untested

Decoherence — Zurek and Zeh, 1970s onward — explains why we never *see* superposed pointers. A macroscopic device entangles with $\sim 10^{23}$ environmental degrees of freedom in femtoseconds, and interference between branches becomes unobservable in practice. This is solid, calculable physics and it dissolves the *appearance* problem. It does not solve the measurement problem: it explains why branches stop interfering, not why one is selected. Conflating the two is the most common error in the literature you will encounter.

Analogy boundary — decoherence as information leakage. Decoherence is genuinely information leakage: which-path data escapes into the environment, which is why it is irreversible in the same statistical sense as your audit chain — reversal is not forbidden, only overwhelmingly improbable. The boundary: leakage explains *loss of interference*, never *outcome selection*. In your pipeline, a state machine that stops being able to backtrack still ends in one state because it was always in one state. A decohered quantum system has no such prior fact — that is precisely what remains unexplained.

The Copenhagen interpretation

Bohr and Heisenberg's position, roughly: the formalism predicts measurement outcomes, and asking what happens between measurements is not a well-posed question. Bohr's **complementarity** holds that wave and particle descriptions are both necessary and mutually exclusive; which applies depends on the experimental arrangement, and no single picture covers both.

Einstein objected for two decades — not to the predictions, which he never disputed, but to the claim of completeness. "God does not play dice" is remembered; the substantive objection came in 1935 with EPR, and is Week 4's material.

The engineering assessment: Copenhagen is an interface specification with the implementation deliberately unspecified. That is defensible for a working programmer and unsatisfying for an architect, which is roughly how the field has been divided ever since.

Key takeaways

- Matrix mechanics and wave mechanics are the same theory in different bases; Dirac notation is the basis-agnostic form.
- $[\hat{x}, \hat{p}] = i\hbar$ implies no shared eigenbasis for position and momentum. Uncertainty is a property of states, not of instruments.
- The Born rule makes $|\psi|^2$ a probability density, splitting the theory into deterministic unitary evolution and probabilistic measurement.
- The measurement problem is the inconsistency of completeness, universal unitarity, and definite outcomes. Interpretations differ in which they abandon.
- Decoherence explains the absence of visible macroscopic superposition. It does not explain outcome selection, and the distinction is load-bearing.

Self-check

Q1 (Objective 2) — Someone tells you decoherence solved the measurement problem in the 1980s. Identify the specific step their account is missing, and give the analogous gap in a distributed-systems consensus protocol that reaches agreement but cannot say which value was proposed first.

Q2 (Objective 3) — Explain why non-commutation is a stronger claim than "measuring one property disturbs the other," and name the experiment that distinguishes the two readings.

Check your answers

A1 — The missing step is **outcome selection**. Decoherence shows the reduced density matrix of the system becomes diagonal in the pointer basis — interference terms are suppressed below any observable threshold. This is a rigorous result and explains why you never see a superposed pointer.

But a diagonal density matrix describes an *ensemble* of possible outcomes with classical probabilities. It does not describe one outcome occurring. Formally, decoherence takes you from a coherent superposition to an improper mixture; the transition from improper mixture to a single actualized result is not derived — it is assumed. Everett's response is to deny that any selection happens; Copenhagen's is to say the question is malformed. Neither is *decoherence*, which is a mechanism, not an interpretation, and is common to all of them.

The consensus analogy: a protocol that guarantees all nodes converge on one value has solved *agreement*, not *ordering*. Every node reports the same answer, and no node can reconstruct which proposal arrived first — the information is genuinely destroyed. Decoherence guarantees the branches stop talking to each other. It says nothing about which branch you find yourself in, and there is no fact in the formalism to appeal to.

A2 — The disturbance reading says each observable has a definite value, and the apparatus perturbs it — an epistemic limit on a system with well-defined properties. Non-commutation says something ontologically stronger: **no state assigns definite values to both simultaneously**, no matter how gently you probe. There is nothing to disturb.

The distinguishing experiments are **Bell tests** (Week 4). If particles carried definite pre-existing values for all measurable quantities, correlations across separated measurements would satisfy Bell's inequality regardless of disturbance — the bound follows from the mere existence of a joint distribution over outcomes, not from any assumption about apparatus. Experiments violate the inequality, and the loophole-free tests of 2015 (Delft, NIST, Vienna) closed the locality and detection gaps simultaneously. The 2022 Nobel to Aspect, Clauser, and Zeilinger recognized this line of work.

The refined statement: quantum mechanics does not merely limit what you can *learn* about simultaneous values. It denies that the values exist to be learned.